Smallest Connected Polyiamond Containing All Free n-Iamonds

a(n) = minimum triangular cells in a connected polyiamond that contains every free n-iamond as a sub-pattern (under translation, rotation, reflection).

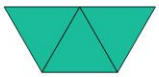
a(1..14) = 1, 2, 3, 5, 6, 9, 12, 16, 19, 23, 27, 31, 36, 41

Computed by Peter Exley, March 2026

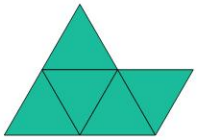
$a(1) = 1$ **[PROVED]** 1 cells, 1 free pieces, bbox 1x1, solved in 0.0s

$a(2) = 2$ **[PROVED]** 2 cells, 1 free pieces, bbox 1x2, solved in 0.0s

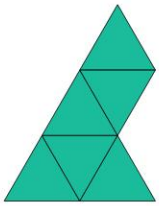
$a(3) = 3$ **[PROVED]** 3 cells, 1 free pieces, bbox 1x3, solved in 0.0s



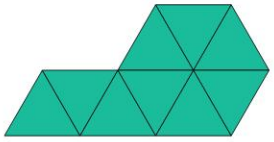
$a(4) = 5$ **[PROVED]** 5 cells, 3 free pieces, bbox 2x4, solved in 0.0s



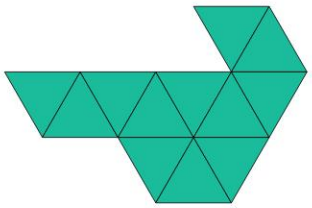
$a(5) = 6$ **[PROVED]** 6 cells, 4 free pieces, bbox 3x3, solved in 0.0s



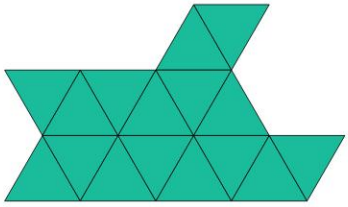
$a(6) = 9$ **[PROVED]** 9 cells, 12 free pieces, bbox 2x6, solved in 0.0s



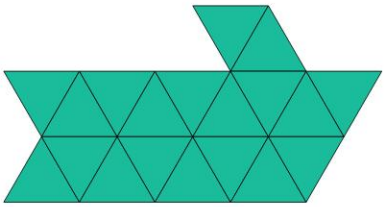
$a(7) = 12$ **[PROVED]** 12 cells, 24 free pieces, bbox 3x7, solved in 0.1s



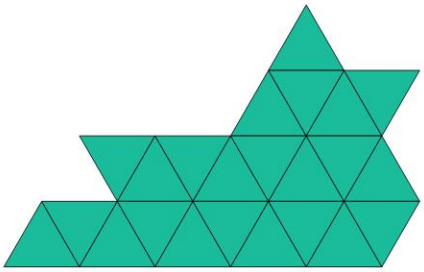
$a(8) = 16$ **[PROVED]** 16 cells, 66 free pieces, bbox 3x8, solved in 0.3s



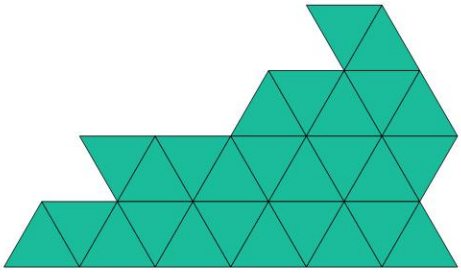
$a(9) = 19$ **[PROVED]** 19 cells, 160 free pieces, bbox 3x9, solved in 0.9s



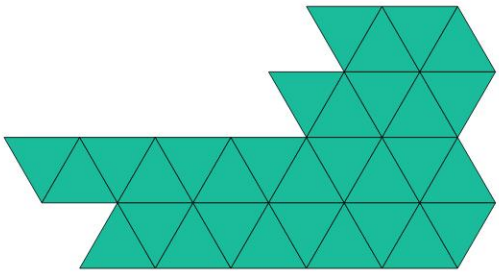
$a(10) = 23$ **[PROVED]** 23 cells, 448 free pieces, bbox 4x10, solved in 3.2s



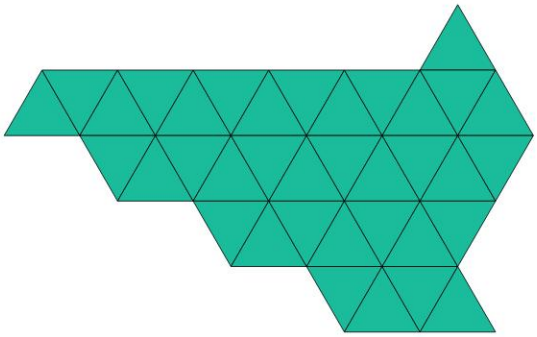
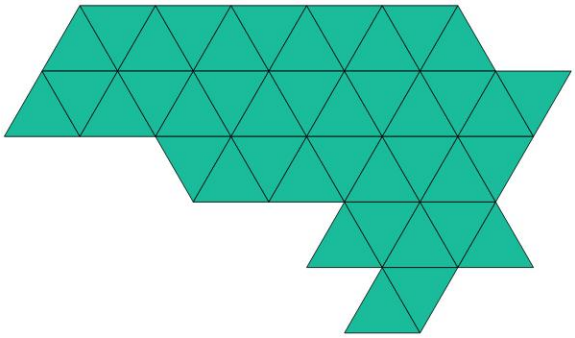
$a(11) = 27$ **[PROVED]** 27 cells, 1186 free pieces, bbox 4x11, solved in 7.2s



$a(12) = 31$ **[PROVED]** 31 cells, 3334 free pieces, bbox 4x12, solved in 25.0s



$a(13) = 36$ **[PROVED]** 36 cells, 9235 free pieces, bbox 5x13, solved in 254.9s



$a(14) = 41$ **[PROVED]** 41 cells, 26166 free pieces, bbox 5x14, solved in 831.5s